Algebraic Topology: Selected Problems & Solutions

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This document contains rigorous solutions and analytical write-ups for various foundational problems in point-set and algebraic topology, focusing on homotopy equivalences, retractions, locally connected spaces, fundamental groups, and deformation retracts.

"These solutions are not originally mine—I used Math Stack Exchange and Discord."

Question: Let A and X be topological spaces, and let $i : A \to X$ be a homotopy equivalence. Does this imply that A is a retract of X?

Solution: No. Consider the inclusion $i : (0,1) \hookrightarrow [0,1]$, which is a homotopy equivalence. However, $(0,1)$ is not a retract of $[0,1]$.

Question: Let X be a topological space. Suppose that for every $x \in X$ and every open neighborhood U of x, there exists a connected neighborhood V of x such that $x \in V \subseteq U$. Show that local connectedness is preserved under retractions, and verify whether $X = I \times \{0\} \cup A \times I$ (where $A = \{0, 1, \frac{1}{2}, \dots\}$) is locally connected.

Solution: Local connectedness is preserved under retractions since continuous maps preserve connectedness and open/neighborhood properties behave well under retractions. However, the space $X = I \times \{0\} \cup A \times I$ is not locally connected.

Question: Let $X = I \times \{0\} \cup A \times I$, where $A = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$. Does there exist a retraction $r : I \times I \to X$?

Solution: No such retraction r exists, as it would violate properties of local connectedness (since $I \times I$ is locally connected while X is not).

Question: Does a local homeomorphism necessarily imply a covering map?

Solution:

Question: Find a space X and base points such that $\pi_1(X, x_0) \neq \pi_1(X, \tilde{x}_0)$.

Solution: Let $X = \mathbb{R} \times S^1$ and $x_0 = (0, e)$.

F. Problem 6 (Strong Deformation)

Question: Let $L = \bigcup_{n=0}^{\infty} L_n$, where $L_0 = I \times \{0\}$ and $L_n = \left\{\left(x, \frac{1}{n}\right) \mid x \in [0,1]\right\}$ for $n \in \mathbb{N}$. Does there exist a strong deformation $F : L \times I \to L$ satisfying:

$$\begin{aligned} F(s,0) &= s \\ F(s,1) &= (1,0) \\ F((1,0),t) &= (1,0) \quad \forall t \in I? \end{aligned}$$

Proof by Contradiction: Assume such a strong deformation F exists. Consider the sequence $p_n = \left(1, \frac{1}{n}\right)$ for $n \in \mathbb{N}$, and define the path $\alpha_n := F|_{p_n \times I}$ from $\left(1, \frac{1}{n}\right)$ to $(1,0)$.

Define a mapping $S : L \setminus \{(0,0)\} \to \mathbb{R}$ by:

$$(x, y) \mapsto \frac{y}{x}$$

The image of $L \setminus \{(0,0)\}$ under S is $\left\{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$.

Since I is path-connected, the image set $S(\alpha_n(I))$ must be a single point $\frac{1}{k}$ for some k. Consequently, endpoints $\alpha_n(0) = \left(1, \frac{1}{n}\right)$ and $\alpha_n(1) = (1,0)$ cannot be maintained simultaneously without passing through the origin, meaning there exists $t_n \in I$ such that $\alpha_n(t_n) = (0,0)$.

Taking a convergent subsequence of $(p_n, t_n)_{n\in\mathbb{N}}$:

$$(p_{n_k}, t_{n_k}) \to ((1,0), t)$$

implies $F(p_{n_k}, t_{n_k}) \to H((1,0), t)$. Because $(0,0) \neq (1,0)$, we arrive at a contradiction.
Conclusion: No such strong deformation exists.

Question: Does a strong deformation retract imply a deformation retract? What about the converse?

Solution: Yes, every strong deformation retract is a deformation retract by definition (dropping the fixed-point condition on the subspace A). The converse is not true; there exist deformation retracts that are not strong deformation retracts.

Explicit Construction for Deformation Retracts: Consider Problem 6 and the claim that L is a deformation retract onto the point $(1,0)$.

Let $F : L \times I \to L$ be defined by $F(s,t) = (1-t)s$, and let $G : L \times I \to L$ be defined by $G(s,t) = t(1,0)$. We construct the homotopy $H : L \times I \to L$ given by:

$$H(s,t) = \begin{cases} F(s,2t) & \text{if } 0 \leq t \leq \frac{1}{2}, \\ G(s,2t-1) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$$